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QU**EPFL**

QUANTUM ICING

Federico Quartieri, Alessandro Ruzza,
Gabriele Santandrea, Daniele Salvi

Q**PFL** HACKATON 2026



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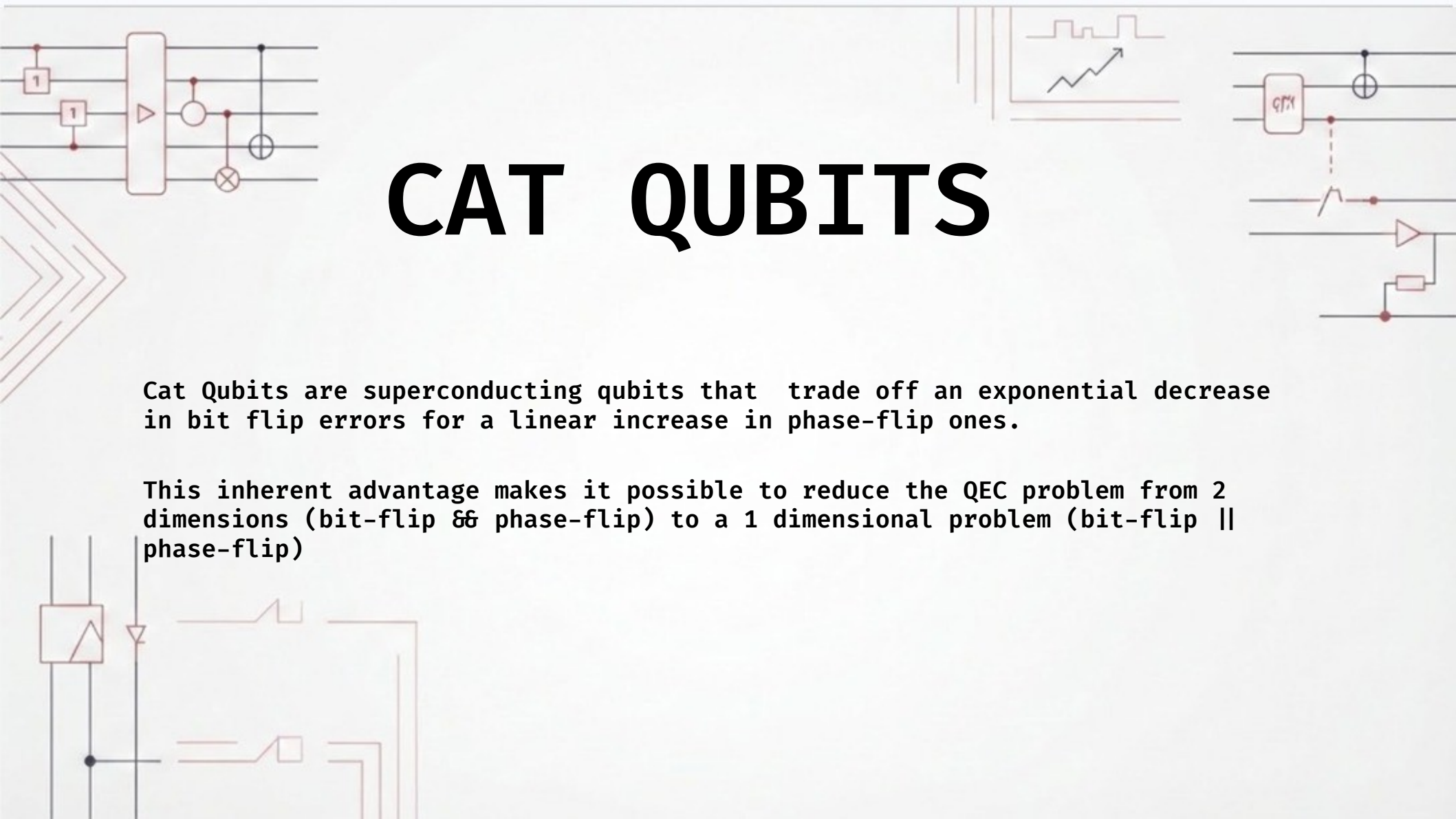
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BIT

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Lausanne, Switzerland 2026



CAT QUBITS

Cat Qubits are superconducting qubits that trade off an exponential decrease in bit flip errors for a linear increase in phase-flip ones.

This inherent advantage makes it possible to reduce the QEC problem from 2 dimensions (bit-flip & phase-flip) to a 1 dimensional problem (bit-flip || phase-flip)

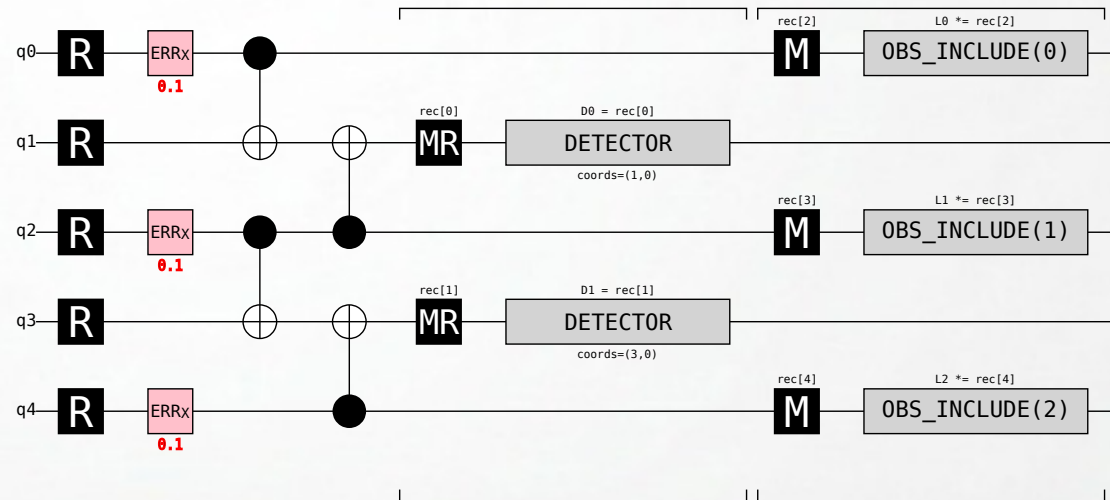
HOW DID WE USE AI?

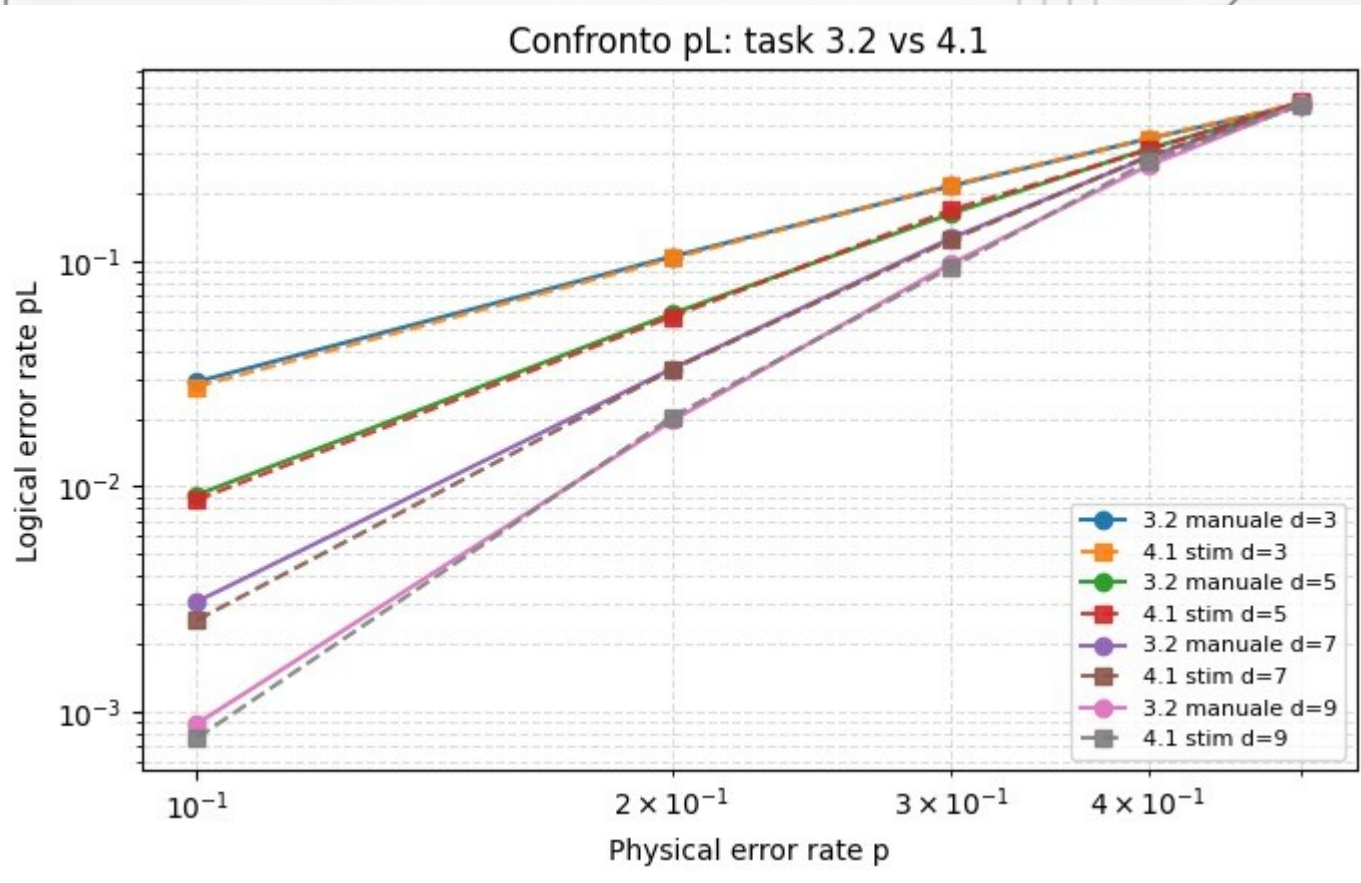


Task 3: Repetition Code and Decoder

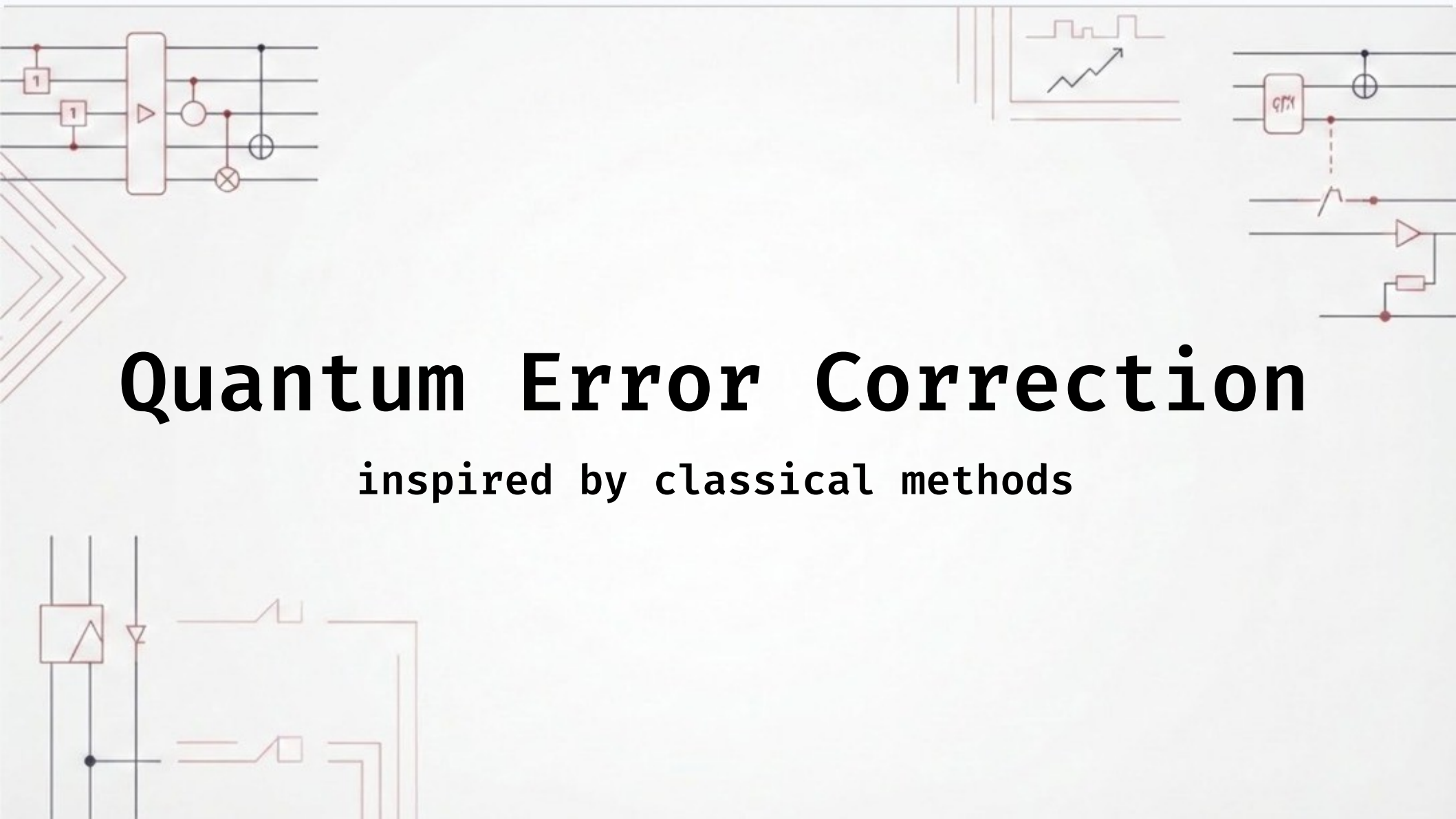
The three ingredients:

- an encoding tuple (n, k, d)
- stabilizers
- decoder





Our manual simulation matched stim's circuit

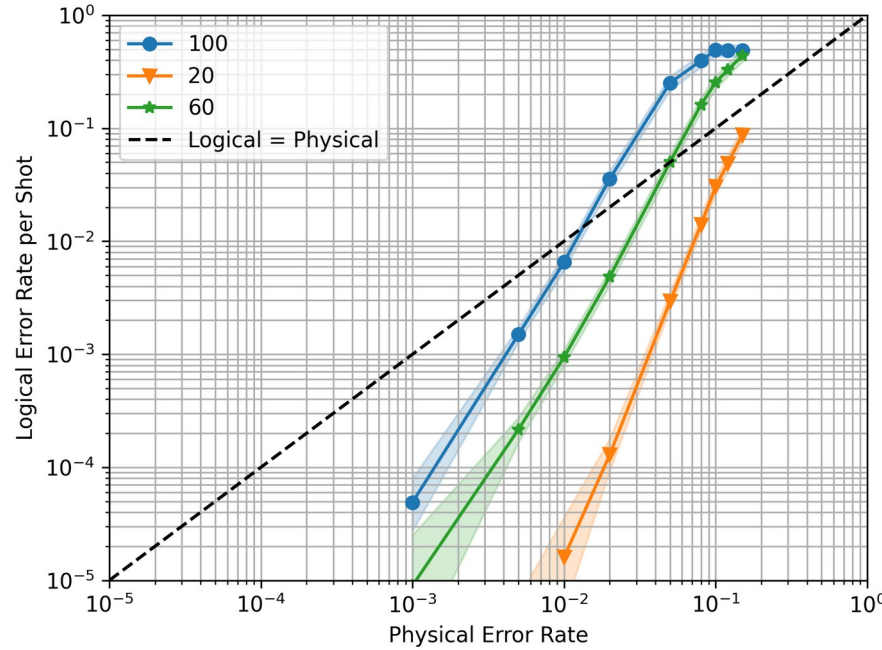


Quantum Error Correction

inspired by classical methods

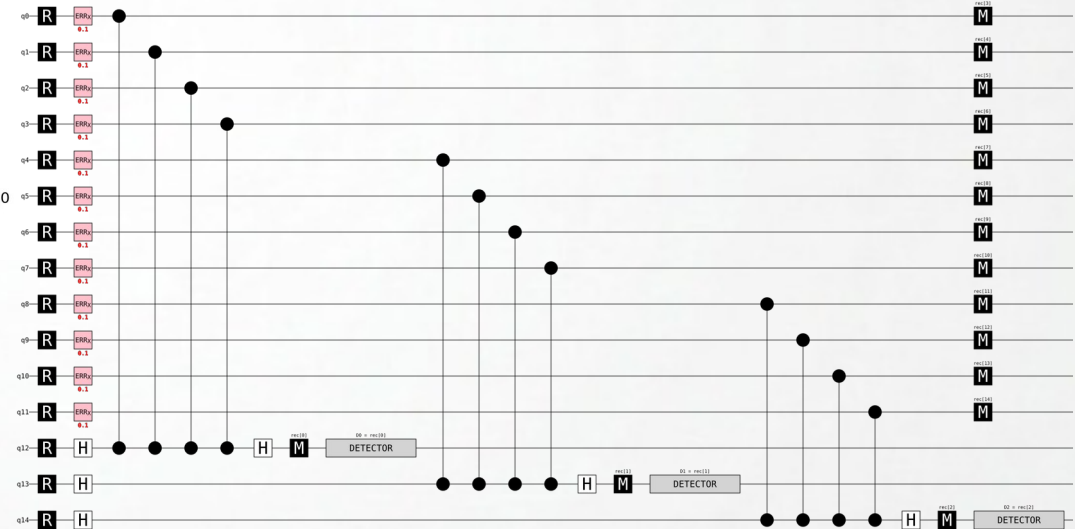
Low-Density Parity Codes

A brave experiment



AN INTUITIVE IMPLEMENTATION OF LDPC

DECODER (BP-OSD):
Belief Propagation + Ordered Statistics Decoding



Custom H

(made by us, 2 forms)

Format A (1 ancilla per 4 data + 2 global)

1	1	1	1	0	0	0	0	Ancilla #1
0	0	0	0	1	1	1	1	Ancilla #2
1	1	0	0	1	1	0	0	Global #1
1	0	1	0	1	0	1	0	Global #2

Format B (3 ancillas per 4 data)

1	1	1	1	0	0	0	0	Block #1
1	1	0	0	0	0	0	0	
1	0	1	0	0	0	0	0	
0	0	0	0	1	1	1	1	
0	0	0	0	1	1	0	0	Block #2
0	0	0	0	1	0	1	0	

Easily scales by adding 1 row, 4 column.

Added row (before globals) contains 0s – 1111

Global rows repeat the “1100” / “1010” pattern in the new columns.

Rate grows towards $4/5 \sim 0.8$

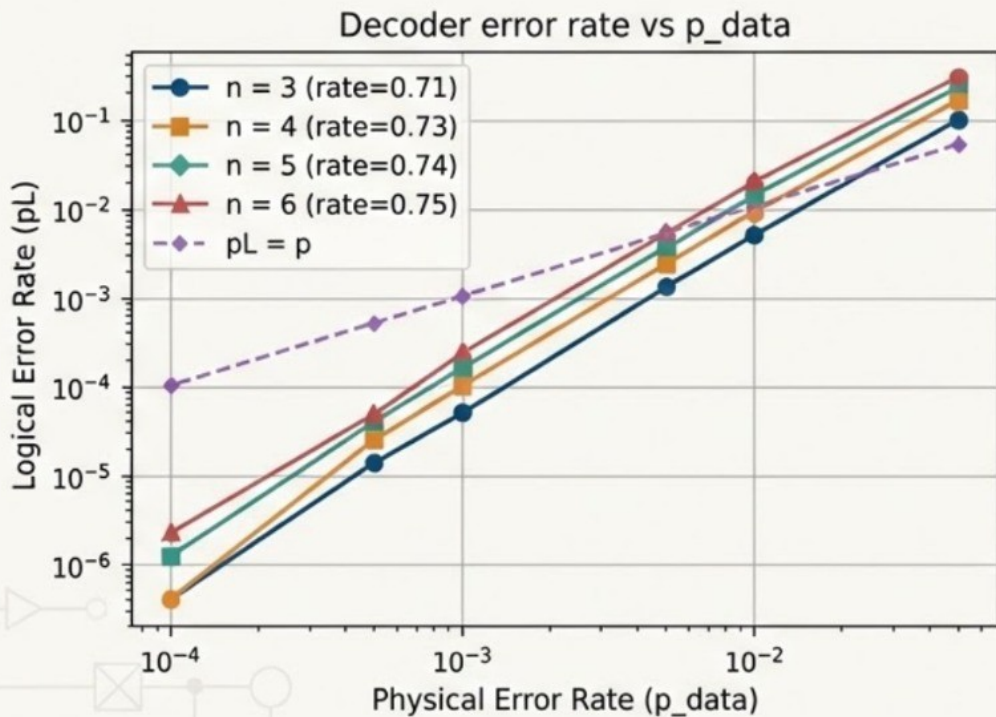
Easily scales by adding 3 rows, 4 columns

Added rows contain the pattern along diagonal.

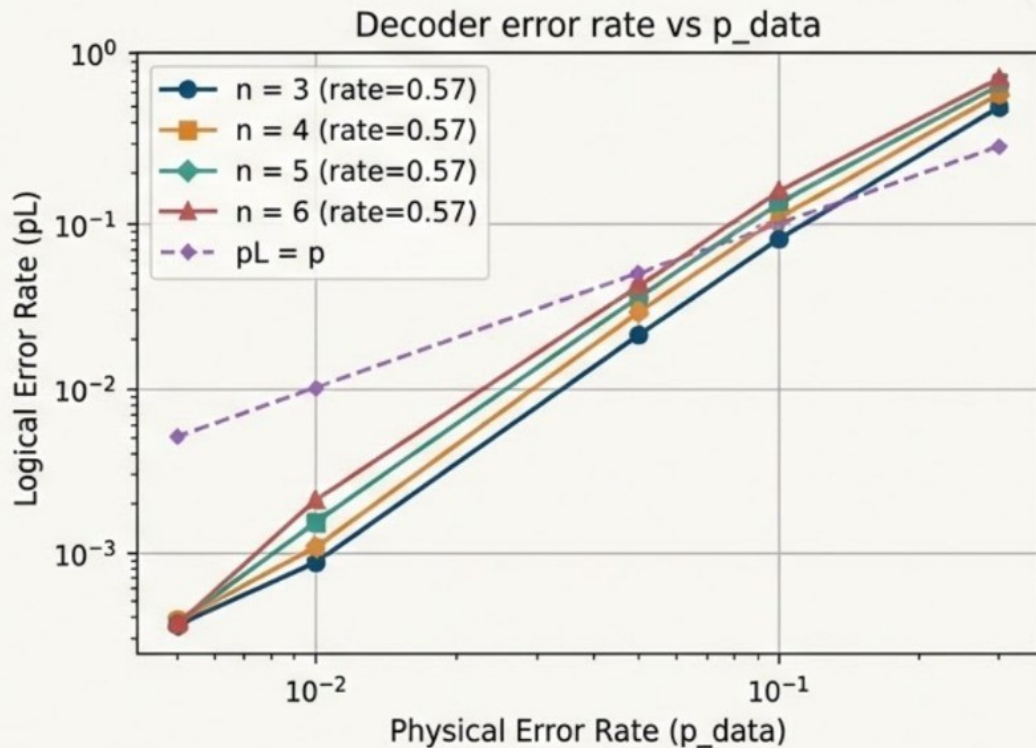
Rate fixed at $4/7 \sim 0.57$

Logical vs Physical Error Rate Comparison

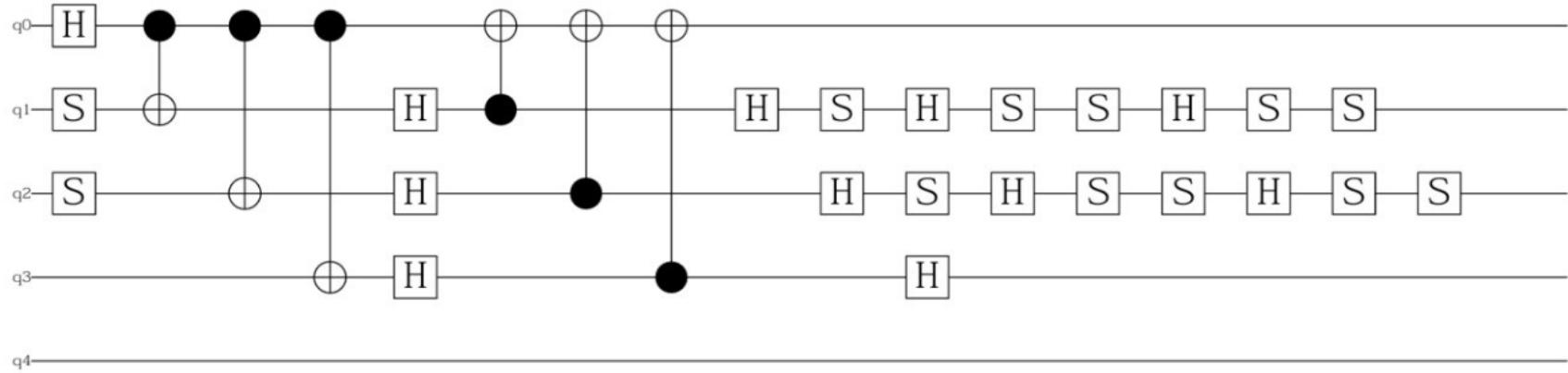
Format A: (1 ancilla/4 data)



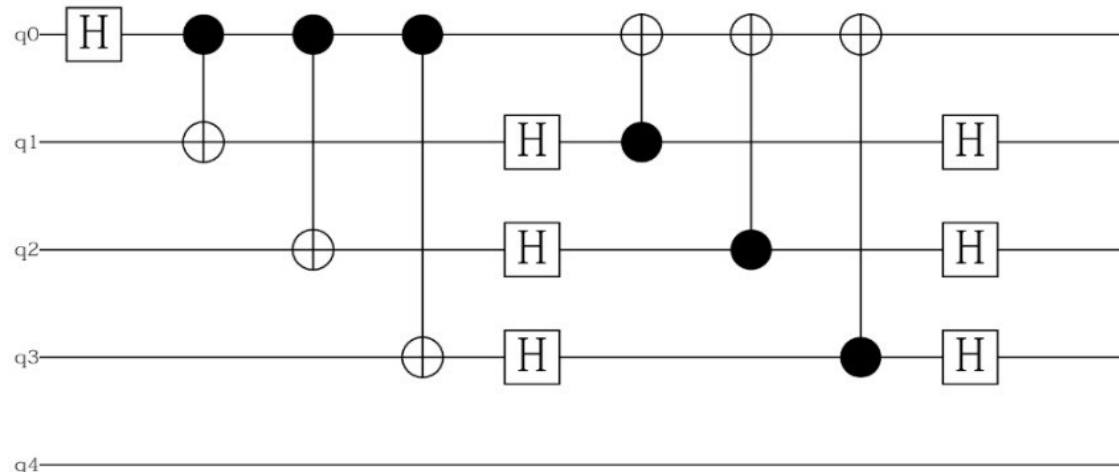
Format B: (3 ancillas/4 data)



Logical HGate implementation A

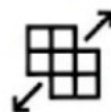


Logical HGate implementation B



Custom H distance increase (made by us, 2 forms)

Format A (1 ancilla per 8 data + 3 global)

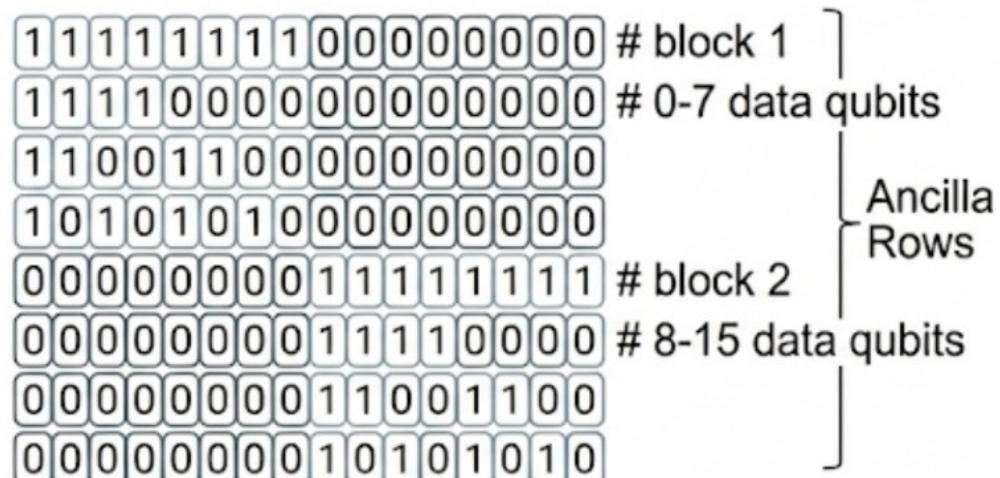


Number of global ancillas scale as $\log(d)$



Rate grows towards $8/9 \sim 0.888$

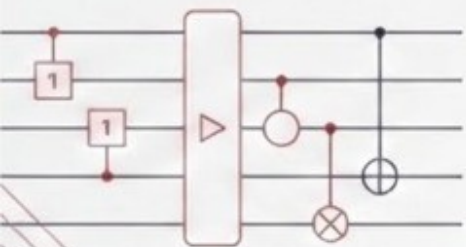
Format B (4 ancillas per 8 data)



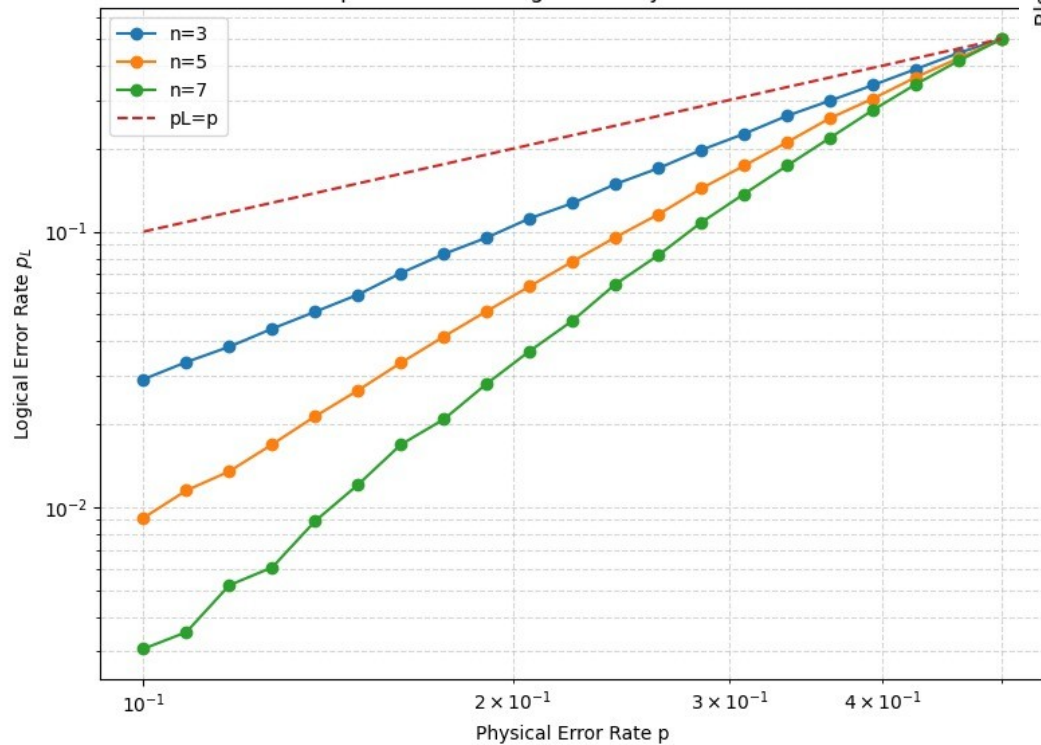
Scales by adding a header row (block) and 3 pattern rows for sets of 8 data qubits. Pattern rows are segmented by data qubit groups.



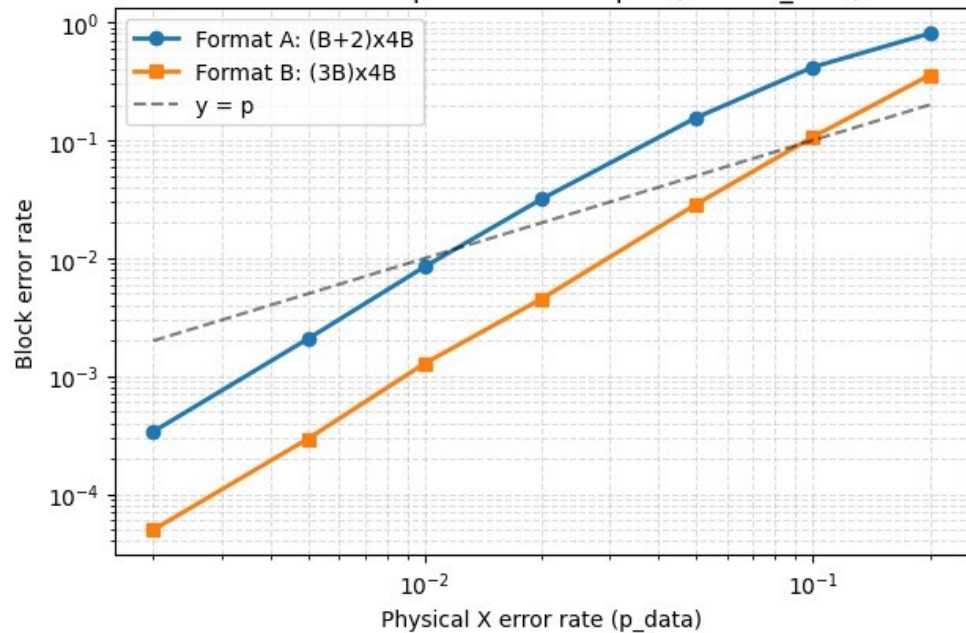
Rate fixed at $8/12 \sim 0.666$



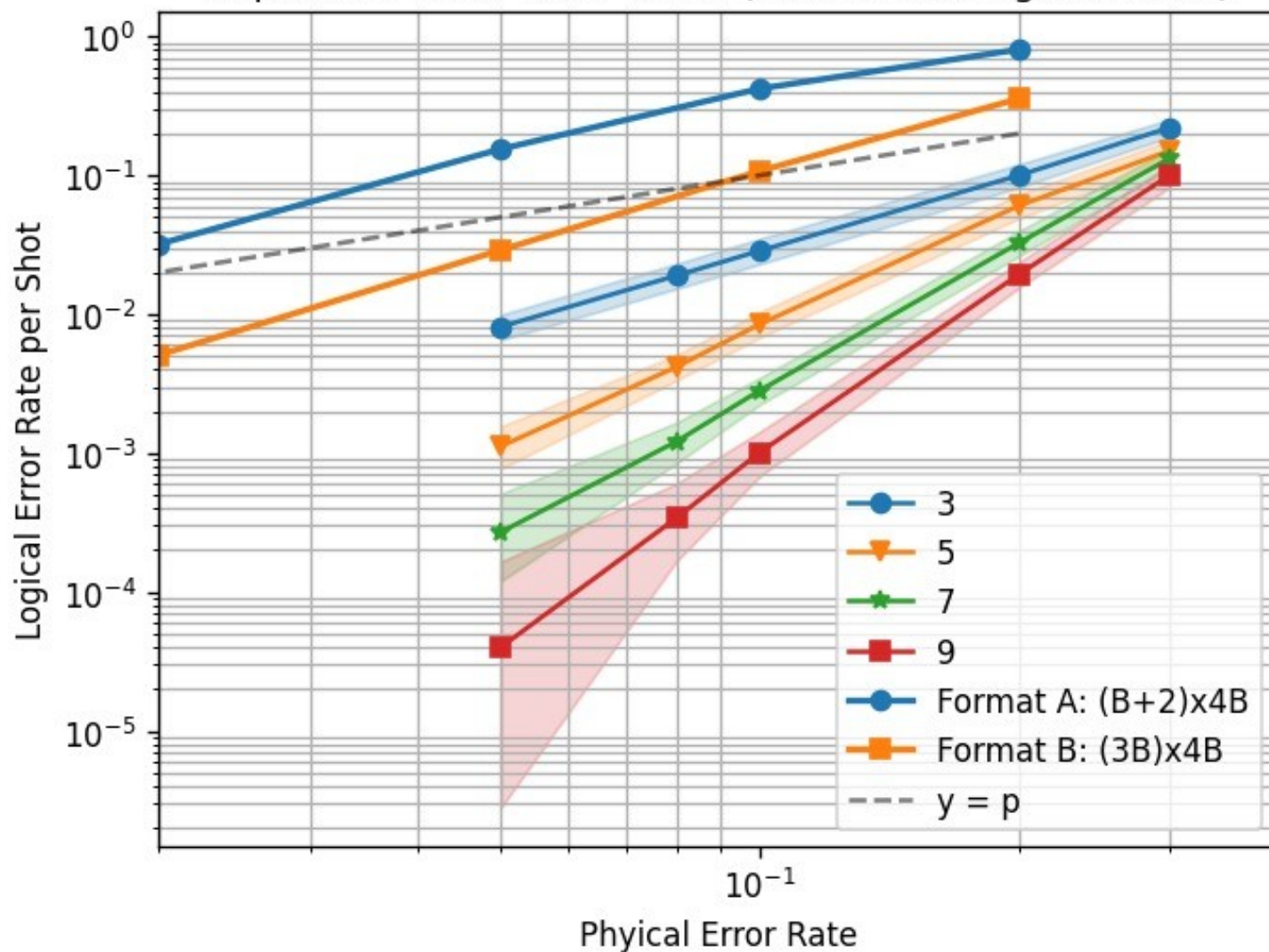
Repetition Code: Logical vs Physical Error Rate



Custom H comparison on one plot (same n_{data})



Repetition Code Error Rates (Phenomenological Noise)



Binary Golay

The *perfect* linear code

Punctured Binary Golay

[n=23, k=12, d=7]-code

$$t = \lfloor \frac{7-1}{2} \rfloor = 3$$

(correct up to 3 bit-flips).

Perfect code: every syndrome maps to a unique error with $wt \leq 3$ (full syndrome table). Block failure for independent bit-flips p :

$$P_{fail}(p) = 1 - \sum_{i=0}^3 \binom{23}{i} p^i (1-p)^{23-i}$$

Reference: Raymond Hill, *A First Course in Coding Theory*, Oxford University Press, 1986.

DECODER

Setup: we use the binary Golay code, which can correct up to $t = 3$ bit-flips error. A bit flip error is modeled as a vector $e \in \{0,1\}^{23}$, where e_i indicates an X error on data qubit i

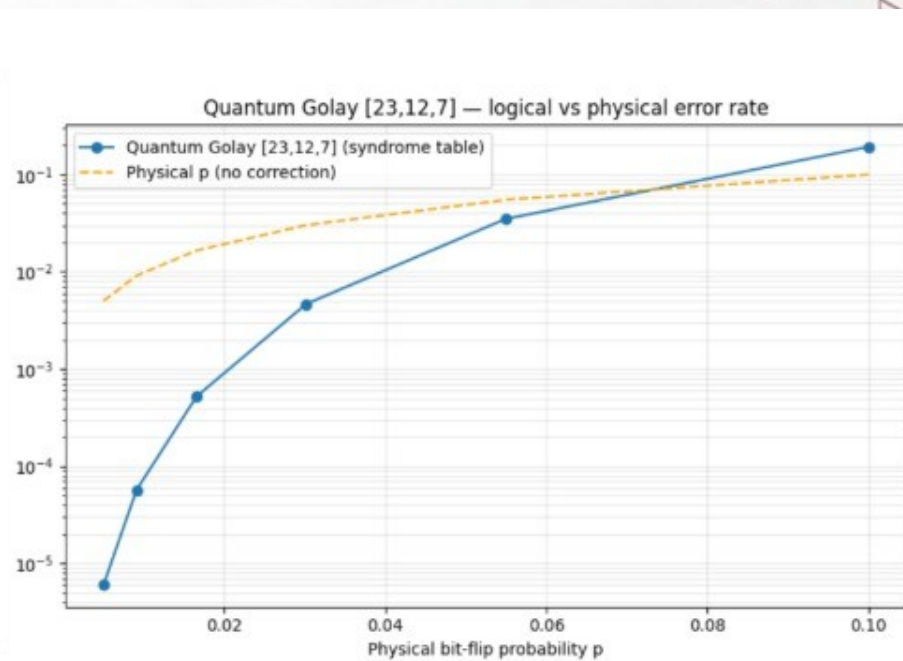
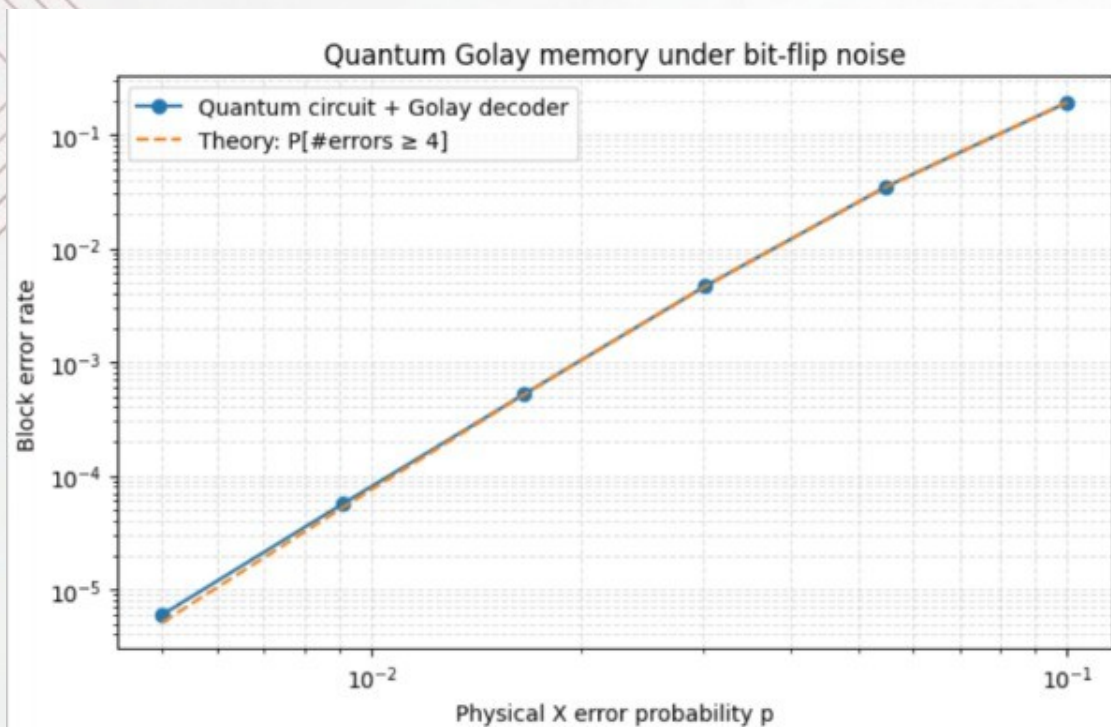
Syndrome extraction: with the parity-check matrix H , the circuit measures 11 parity checks and outputs a syndrome $s \in \{0,1\}^{11}$.

$$s = H * r = H(c \otimes e) = Hc \otimes He = He$$

Lookup table: since the equation $He = s$ generally has multiple solutions, decoding selects the most likely error, which for an independent bit-flip channel corresponds to the minimum-weight solution. We therefore precompute a table T mapping each syndrome to its coset leader $\hat{e} = T(s)$ restricting to errors with $\text{wt}(e) \leq 3$

Correction: $d_{\text{corr}} = d \otimes \hat{e}$

Results



Extended Binary Golay

[n=24, k=12, d=8]-code

$$t = \lfloor \frac{8-1}{2} \rfloor$$

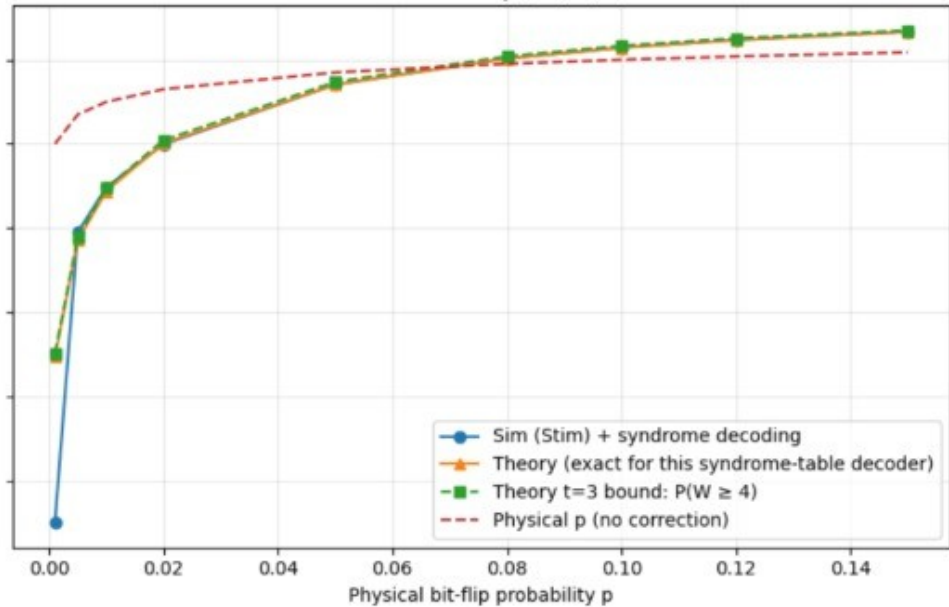
(same bounded-distance correction radius).

Advantage: larger distance (better separation / robustness), cost: +1 qubit per block. Block failure (independent bit-flips p):

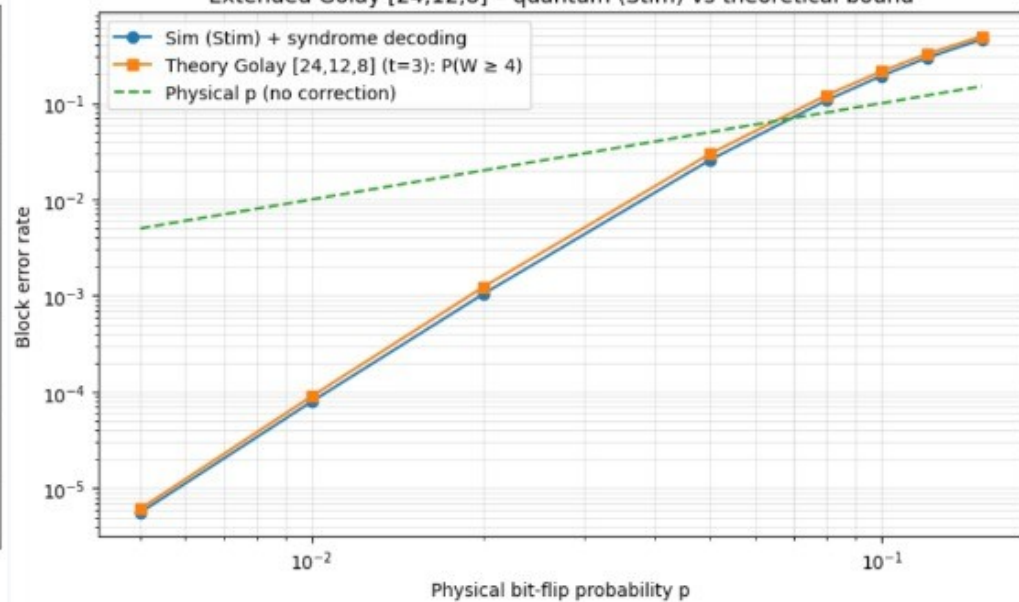
$$P_{fail}(p) = 1 - \sum_{i=0}^3 \binom{24}{i} p^i (1-p)^{24-i}$$

Results

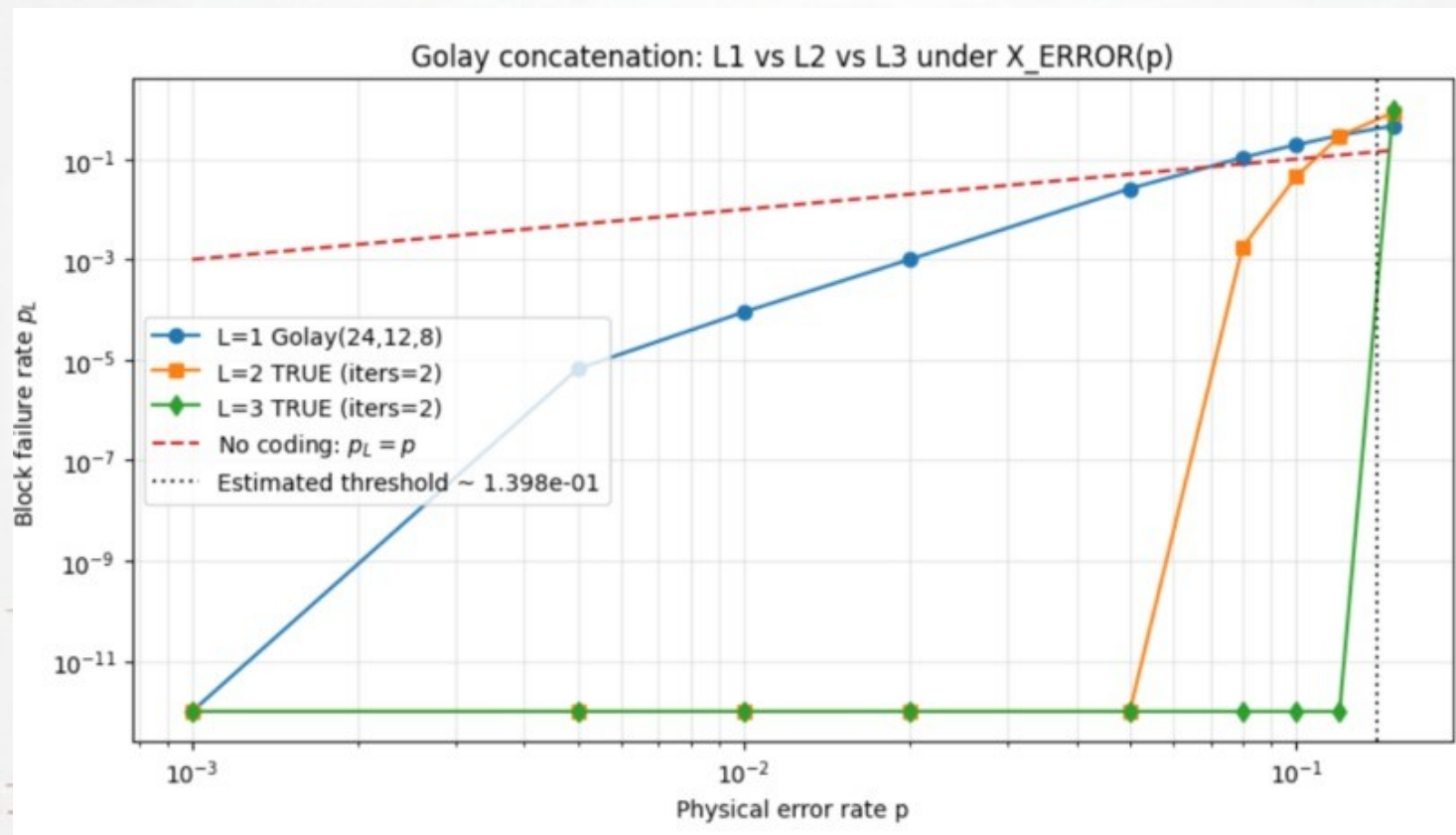
Extended Golay [24,12,8]



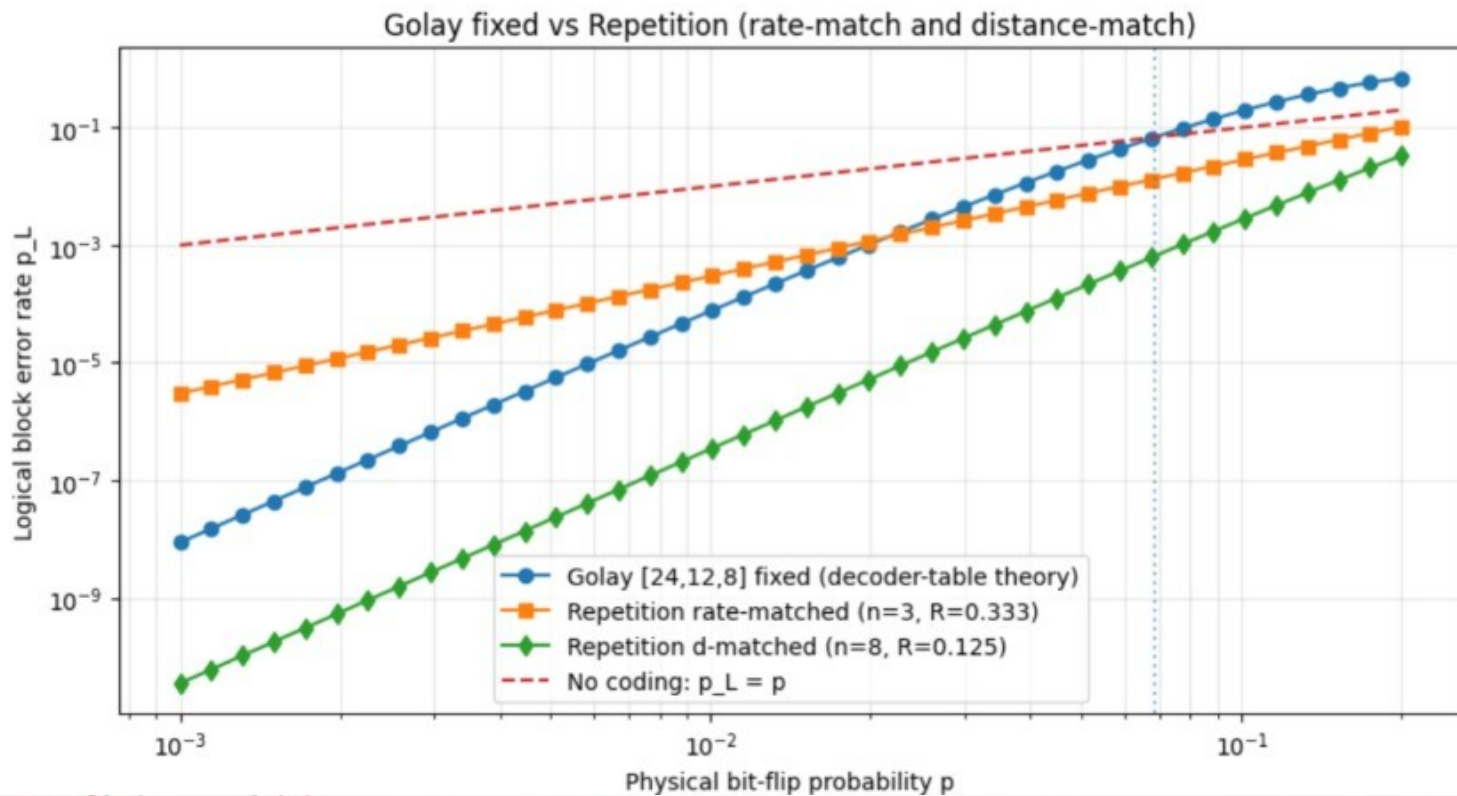
Extended Golay [24,12,8] - quantum (Stim) vs theoretical bound



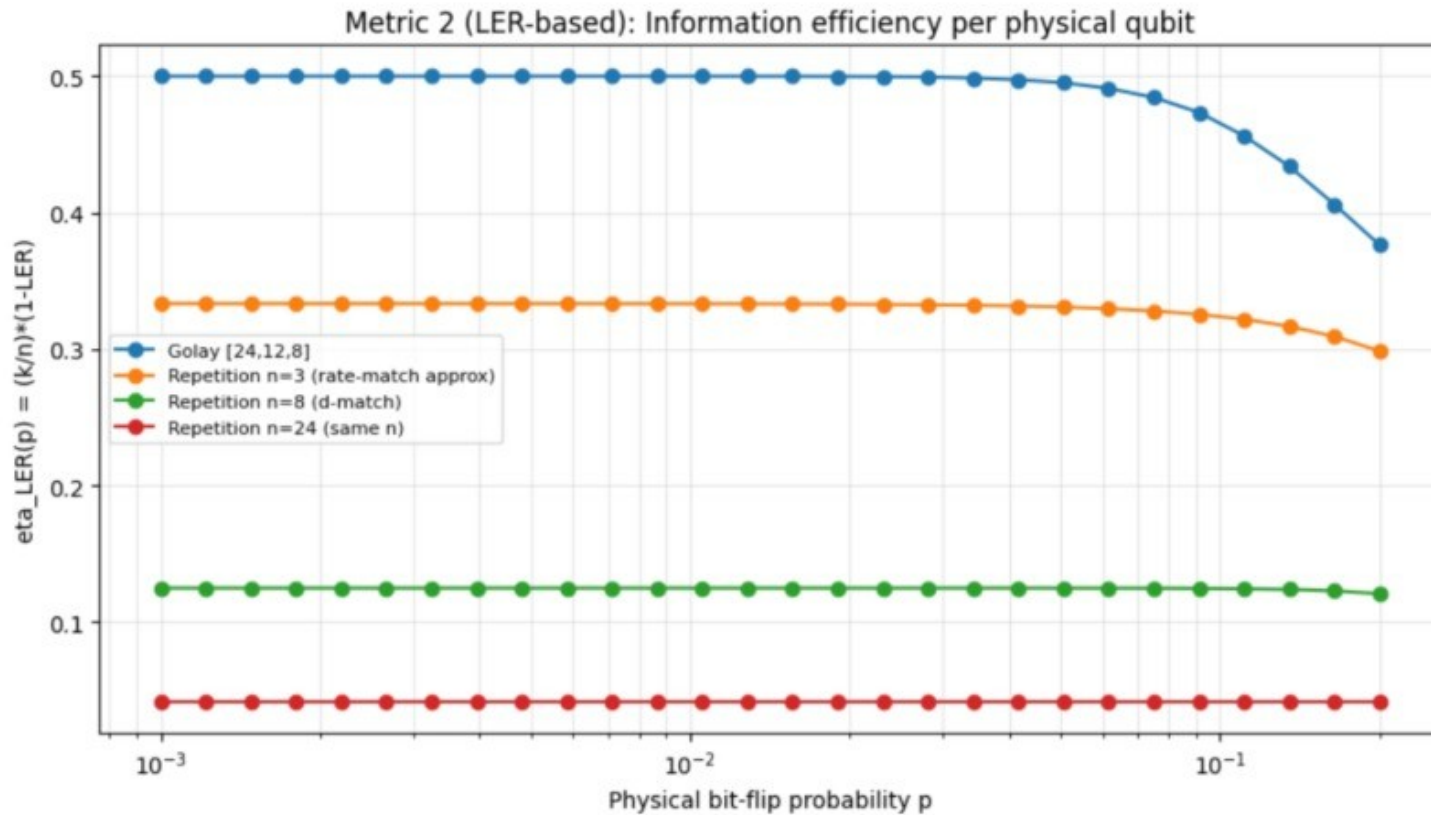
Golay 3 Level concatenation



Golay vs Repetition



Golay vs Repetition



Steane Overview

Parameters: $n=7$, $k=1$, $d=3$

CSS Architecture: Separates X and Z tracking to analyze hardware noise bias.

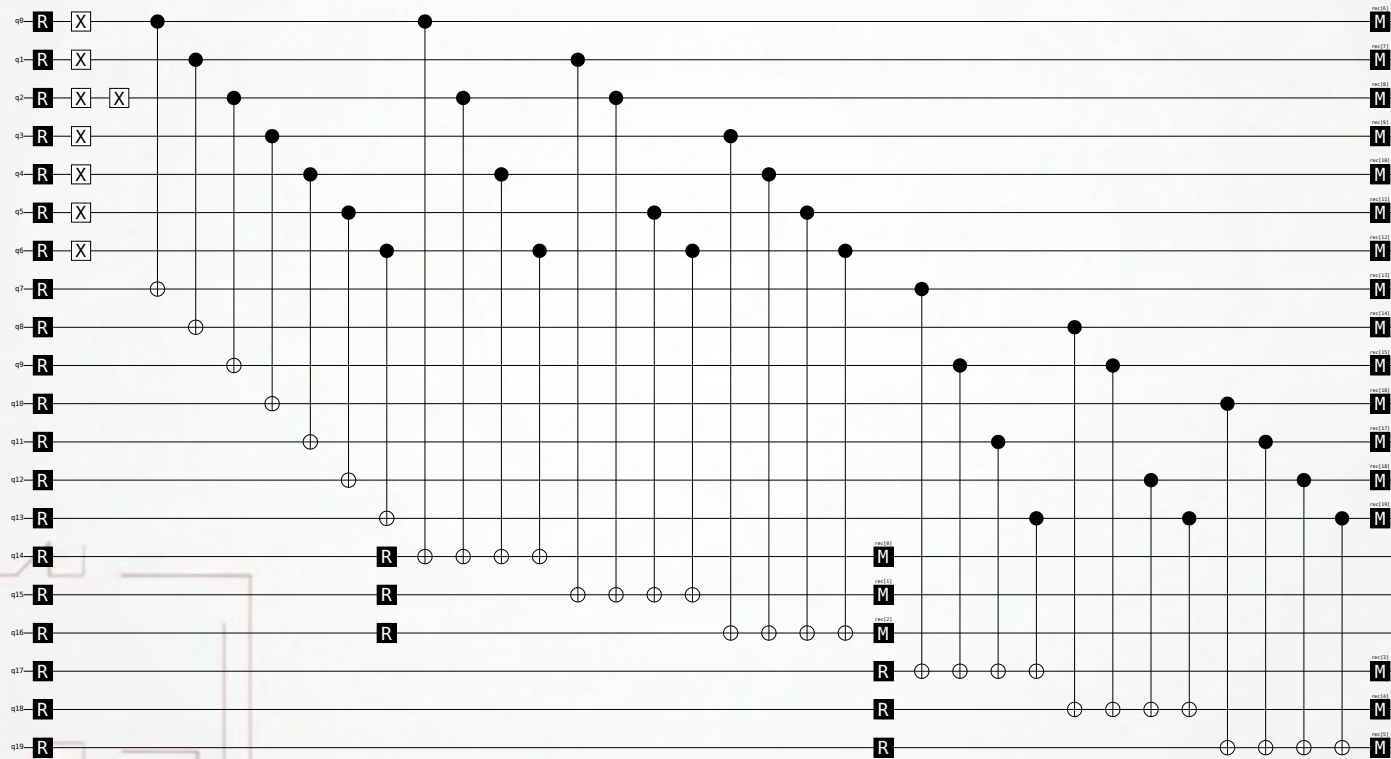
Deterministic Prep: Encodes perfect 0 states without dropping shots.

Safe Extraction: Non-destructive parity checks using dedicated ancilla qubits.

Precise Decoding: Deterministic Hamming-mapped decoder to track physical errors instantly.

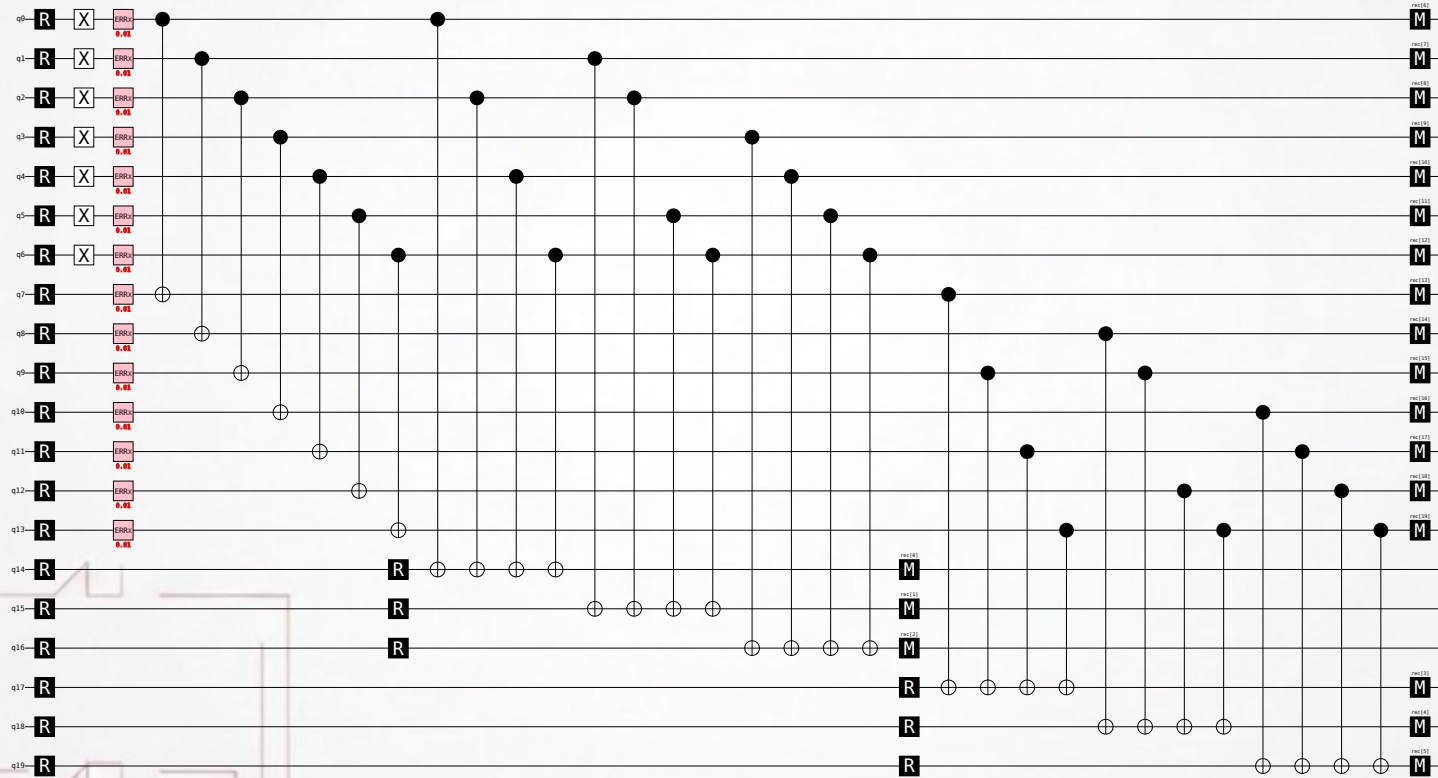
Steane

CNOT - scenario A



Steane

CNOT - scenario B



Steane CNOT overview

The Math: Propagates X to X from control to target.

The Advantage: Never rotates X errors into the lethal Z basis.

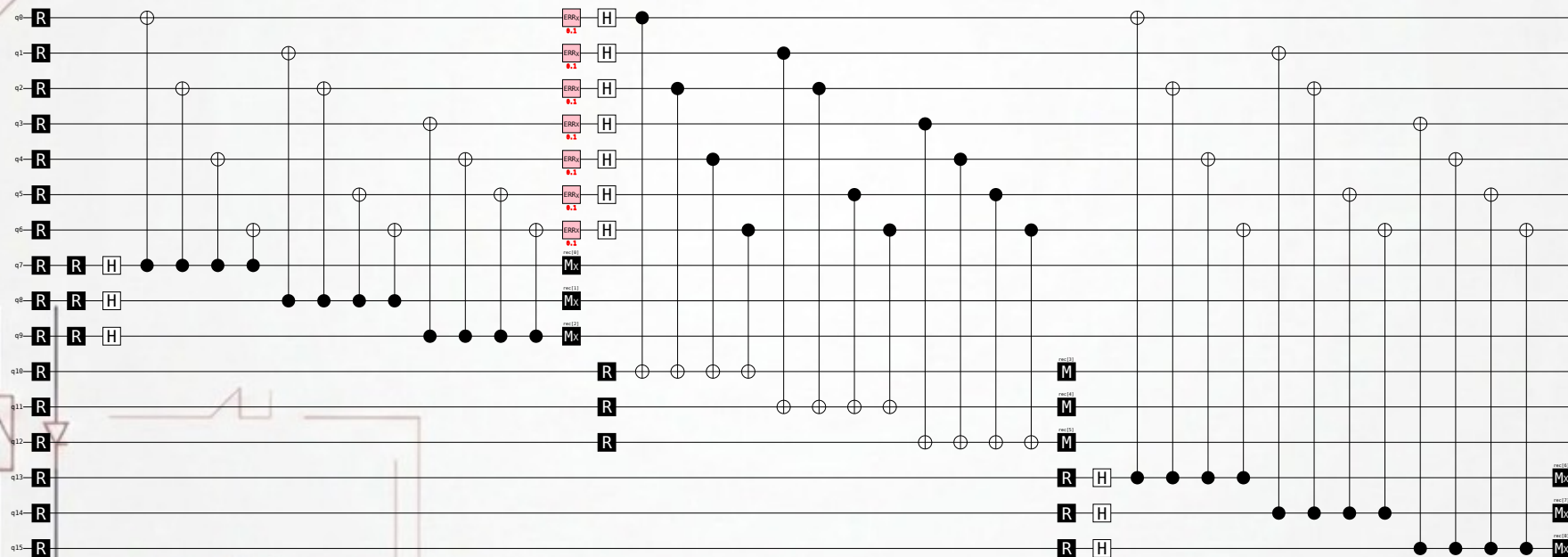
Simulation Error Rate:

- Scenario A: 0 error
- Scenario B: 0.0075 ($p = 0.01$, shots = 20000)

Verdict: CNOT is natively safe. We must use it via **Gate Teleportation** to safely execute H and S gates.

Steane

H - scenario A



Steane

H - scenario B

Steane H overview

Scenario A: Direct Transversal H Gate

~**51.87%** → Z-Error Triggers

The H gate mathematically swaps axes ($X \rightarrow Z$).

Harmless bit flips are converted into lethal phase-flips, completely destroying Cat Qubit bias.

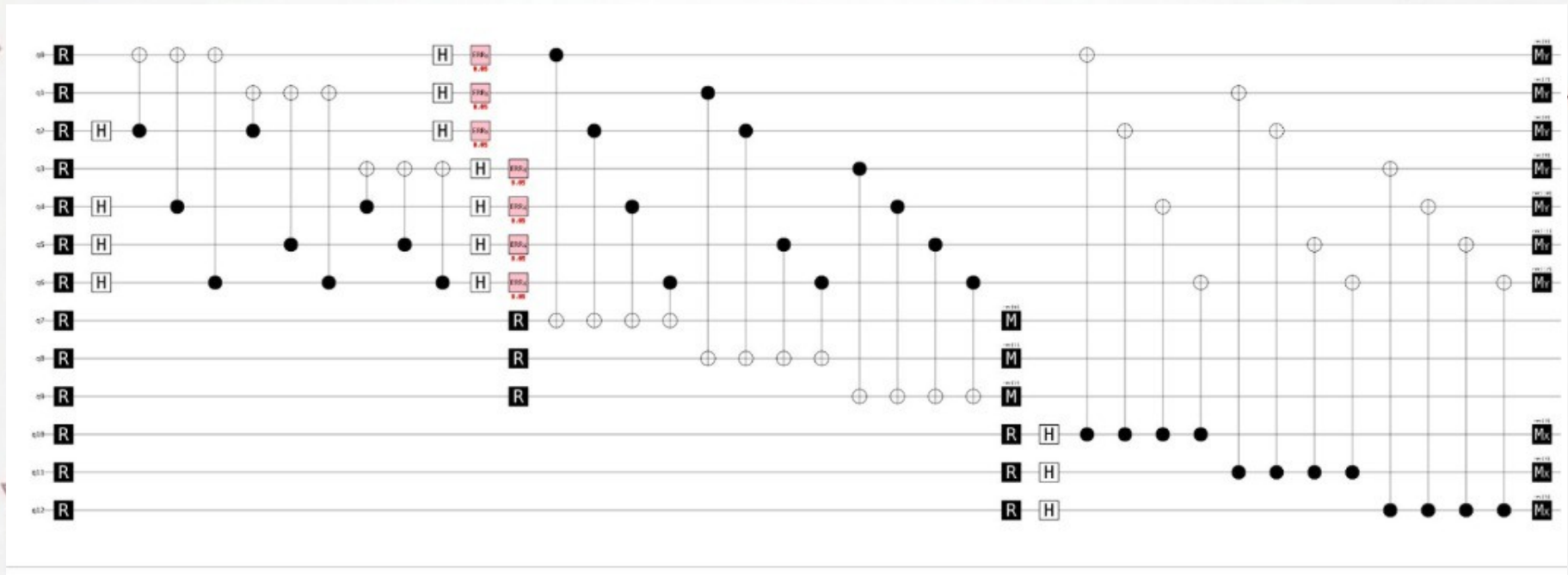
Scenario B: CNOT-Based Gate Teleportation

~**50.15%** → X-Error Triggers

Errors safely propagate strictly as bit-flips ($X \rightarrow X$).

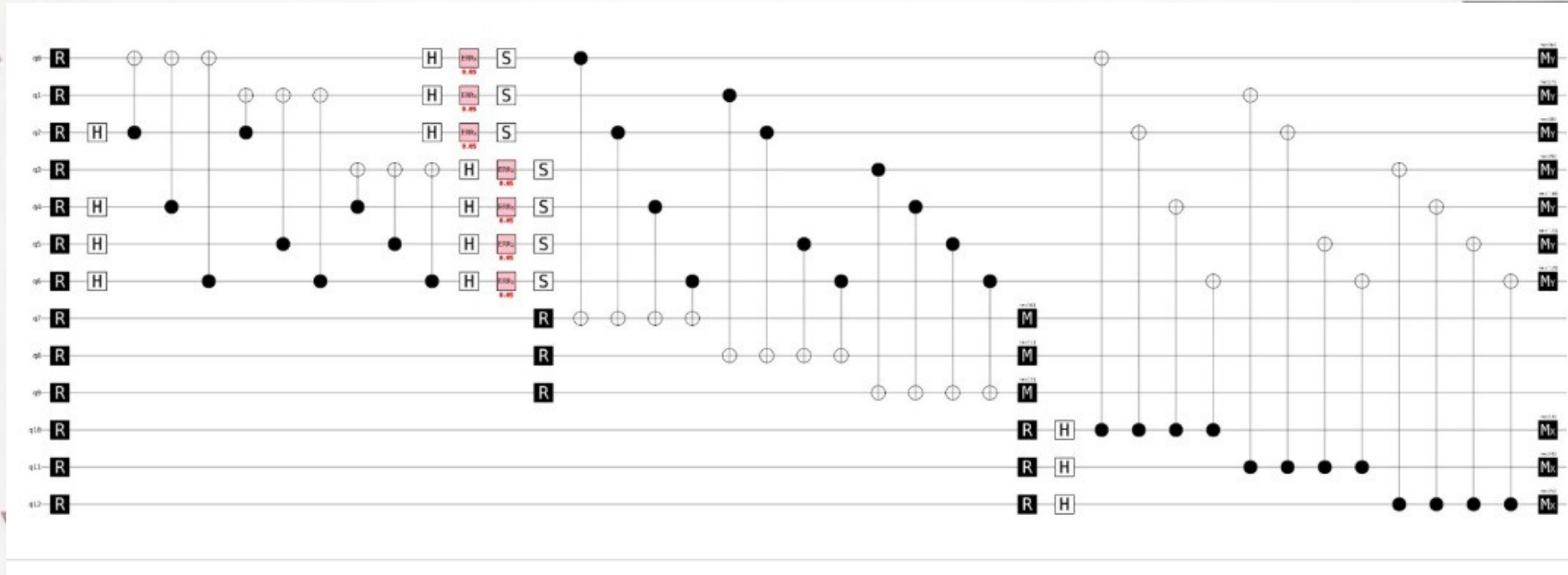
Steane

S - scenario A



Steane

S - Scenario B



Steane S overview

Scenario A: without S Gate

~**30.00%** → X-Error Triggers

0.00% → Z-Error Triggers

Scenario B: with S Gate

~**30.00%** → X-Error Triggers

~**30.00%** → Z-Error Triggers

Golay Gates

Logical meaning: with the common “extended Golay as 12×24 CSS checks” setup you often end up with $k = 0$, so you’re showing stabilizer/syndrome behavior, not a gate on a logical qubit.

Syndrome extraction cost: Golay checks are heavier/denser, so measuring stabilizers needs many more CNOTs + ancillas, making error propagation and scheduling much more important than in small codes.

Syndrome differencing: Golay demos often require reference rounds / syndrome differencing (especially for X-checks), otherwise “syndrome triggers” can be dominated by a nontrivial baseline rather than noise or the gate effect.

Decoding burden: Golay decoding is not a tiny lookup; you typically need a more structured decoder to interpret syndromes reliably.

THANK YOU

Q&A

